

HTIN 5005
Assignment 2

Reproducing Automated ICD Coding on the CodiEsp-D Dataset

Project Code: #17: Automated ICD Coding Using Extreme
Multi-label Long Text Transformer-based Models

Jiayu Xiong / 500037218
Yu Lei / 540649488

1 Introduction

Accurate ICD coding is fundamental to clinical documentation, billing, and epidemiological analysis. Manual coding is slow, costly, and error-prone. With the digitalization of Electronic Health Records, automatic ICD coding has emerged as a key research focus in medical AI. The task involves mapping lengthy, unstructured narratives to thousands of potential diagnosis codes, forming an instance of extreme multi-label long-text classification (XMLTC).

Traditional rule-based and statistical approaches (e.g., MetaMap, cTAKES, SVMs) often fail to capture context or handle label imbalance. Transformer-based deep learning methods have achieved significant progress. Liu et al. (2023) introduced **XR-LAT**, a hierarchical long-text Transformer that leverages label-tree structures and label-aware attention to address data imbalance and document length constraints. This study reproduces their approach using the open-access CodiEsp-D dataset as a linguistically diverse substitute for the restricted MIMIC-III corpus.

2 Dataset

The **CodiEsp-D** corpus comprises de-identified Spanish clinical narratives annotated with ICD-10-ES diagnostic codes. Each record represents a discharge summary containing one or more diagnosis codes derived from the ICD-10-ES taxonomy.

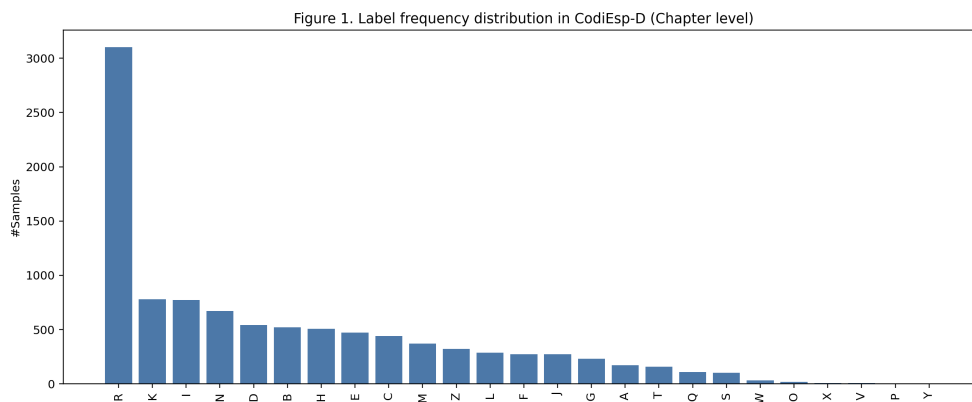


Figure 1: Label frequency distribution in the CodiEsp-D dataset, illustrating strong label imbalance across ICD-10-ES chapters.

As shown in Figure 1, the distribution of labels across ICD-10 chapters is highly skewed. A small number of chapters, such as “R”, account for the majority of annotations—while many chapters contain fewer than 200 samples. This long-tail distribution poses a significant challenge for automatic coding systems, as models tend to overfit to frequent codes and underperform on rare, clinically important diagnoses.

In extreme multi-label settings such as this, class imbalance can distort both optimization and evaluation:

- **During training**, loss gradients are dominated by common codes, leading to poor representation learning for rare labels.
- **During inference**, uniform thresholds or global calibration may favor head classes and suppress low-frequency predictions.
- **During evaluation**, micro-averaged metrics (e.g., micro-F1) are biased toward frequent labels, whereas macro-level metrics provide a fairer view of rare-label performance.

To mitigate this imbalance, later sections introduce techniques such as **Asymmetric Loss (ASL)** for reweighting negative samples and hierarchical supervision to enhance generalization across label levels. The dataset’s multilingual and open-access nature makes it a practical yet challenging benchmark for reproducing automatic ICD coding methods.

3 Methods

We compare two architectures: (1) **PLM-ICD/HiLAT baseline** without hierarchical constraints, and (2) the **XR-LAT** model with hierarchical decoding. Both use the Spanish biomedical RoBERTa encoder (PlanTL-GOB-ES/roberta-base-bne) fine-tuned on CodiEsp-D.

3.1 Text Encoding and Document Representation

Given a clinical document $x = \{t_1, \dots, t_N\}$, we segment it into S overlapping chunks of length C :

$$x = [x^{(1)}, x^{(2)}, \dots, x^{(S)}], \quad C = 256, S = 6.$$

Each chunk is encoded by RoBERTa:

$$H^{(s)} = \text{RoBERTa}(x^{(s)}) \in \mathbb{R}^{C \times d}.$$

All hidden states are concatenated and re-aggregated:

$$H = [H^{(1)}; \dots; H^{(S)}] \in \mathbb{R}^{(SC) \times d}, \quad h = \text{AttnPool}(H).$$

This yields a single document embedding h used for multi-label prediction.

3.2 Label-wise Attention and Hierarchical Decoding

For L possible labels, we define label-specific attention:

$$Z = \tanh(W_1 H), \quad W_1 \in \mathbb{R}^{d \times d}, \quad (1)$$

$$A = \text{softmax}(W_2 Z) \in \mathbb{R}^{L \times (SC)}, \quad (2)$$

$$D = AH, \quad \hat{y} = Dw_c + b_c, \quad w_c \in \mathbb{R}^d. \quad (3)$$

Each label ℓ has a relevance distribution A_ℓ over tokens.

The hierarchical decoding follows the ICD-10 tree:

Chapter $\rightarrow$ Category $\rightarrow$ Code.

Each node v_i has embedding e_i , constrained by parent–child proximity:

$$\|e_i - e_{\text{parent}(i)}\|_2^2 < \epsilon.$$

The multi-level objective is:

$$\mathcal{L} = \lambda_0 \mathcal{L}_{\text{chapter}} + \lambda_1 \mathcal{L}_{\text{category}} + \lambda_2 \mathcal{L}_{\text{code}}, \quad (\lambda_0, \lambda_1, \lambda_2) = (0.2, 0.3, 0.5). \quad (4)$$

Inference optionally applies hierarchical masking:

$$\hat{y}_c = 0 \quad \text{if } \hat{y}_{\text{parent}(c)} < \tau_p.$$

3.3 Training and Calibration

We employ **Asymmetric Loss (ASL)** proposed by Ridnik et al. (2021) to address class imbalance:

$$p_\ell = \sigma(\hat{y}_\ell), \quad q_\ell = 1 - p_\ell, \quad (5)$$

$$\mathcal{L}_{\text{ASL}} = - \sum_{\ell=1}^L \left[y_\ell (1 - p_\ell)^{\gamma_+} \log p_\ell + (1 - y_\ell) (1 - q_\ell)^{\gamma_-} \log q_\ell \right], \quad (6)$$

$$(\gamma_+, \gamma_-) = (0, 4). \quad (7)$$

Optimization uses AdamW with cosine decay and linear warm-up. Hierarchical pretraining proceeds top-down:

Chapter $\Rightarrow$ Category $\Rightarrow$ Code.

Binary predictions are obtained via global threshold τ :

$$\hat{y}_\ell^* = \begin{cases} 1, & \text{if } p_\ell \geq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

Temperature scaling is applied for logit calibration:

$$p'_\ell = \sigma(\hat{y}_\ell/T), \quad T > 0.$$

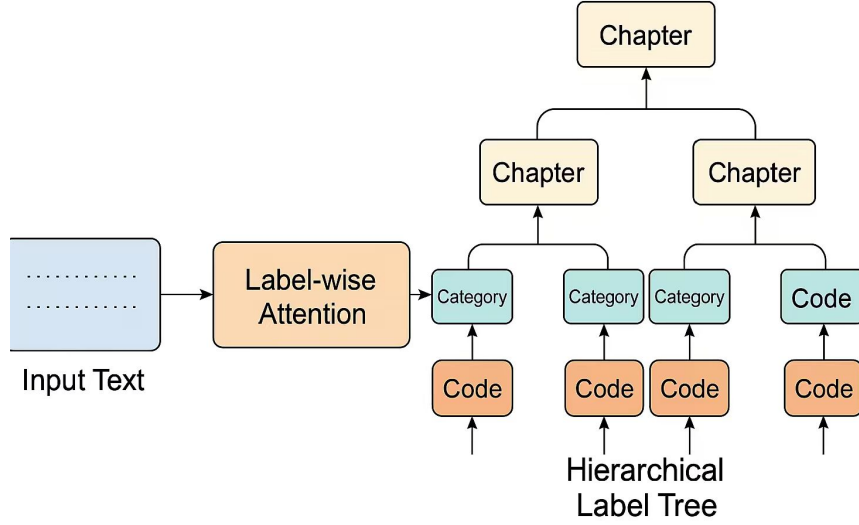


Figure 2: Overview of the XR-LAT architecture. From input text to label-wise attention and hierarchical decoding over ICD-10 levels.

4 Experimental Setup

4.1 Environment

All experiments were conducted on an NVIDIA RTX 4060 GPU (8 GB) using the following software versions:

`transformers 4.19, datasets 2.8, torch 1.11.`

Mixed-precision (FP16) training was enabled to improve memory efficiency and speed.

4.2 Training Configuration

For both the PLM-ICD baseline and the XR-LAT model, we adopt the AdamW optimizer with linear warm-up and cosine decay. Let $\eta_0 = 5 \times 10^{-5}$ denote the initial learning rate, and $T_w = 1000$

the number of warm-up steps. The learning rate at step t follows:

$$\eta(t) = \begin{cases} \eta_0 \cdot \frac{t}{T_w}, & t \leq T_w, \\ \frac{1}{2}\eta_0 \left(1 + \cos \frac{\pi(t - T_w)}{T_{\max} - T_w}\right), & t > T_w. \end{cases}$$

Batch size $b = 2$ with gradient accumulation $n_g = 4$ yields an effective global batch size of $b_{\text{eff}} = 8$. Each model was trained with dropout rate $p = 0.1$, layer normalization, and early stopping based on validation macro-F1.

4.3 Hyperparameters

The baseline was trained for 20 epochs; XR-LAT was trained for 10 epochs per hierarchical level. Both used Asymmetric Loss (ASL) as described in Section 3.3. Key settings are summarized in Table 1.

Table 1: Key hyperparameters for baseline and XR-LAT training.

Parameter	Baseline	XR-LAT (per level)
Chunk length C	256	256
Number of chunks S	6	6
Batch size / Grad accum.	2 / 4	2 / 4
Epochs	20	10
Learning rate η_0	5×10^{-5}	5×10^{-5}
Warm-up steps T_w	1000	1000
Dropout p	0.1	0.1
Loss function	ASL	ASL

4.4 Inference and Threshold Selection

For binary prediction, probabilities $\sigma(\hat{y}_\ell)$ are thresholded using a global τ :

$$\hat{y}_\ell^* = \begin{cases} 1, & \text{if } \sigma(\hat{y}_\ell) \geq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

Grid search over $\tau \in [0.05, 0.9]$ maximizes validation micro-F1. Temperature scaling was applied for logit calibration:

$$p'_\ell = \sigma(\hat{y}_\ell/T), \quad T > 0.$$

This adjustment provided minor improvements in macro-AUC but negligible F1 gains.

5 Results

Unless stated otherwise, all results are reported on the test split using the best validation threshold.

5.1 Code-level Comparison

Table 2: Code-level performance on CodiEsp-D (Test).

Model	micro-F1	micro-Prec	micro-Rec	micro-AUC	macro-F1	macro-AUC
Baseline	0.336	0.322	0.351	0.900	0.046	0.793
XR-LAT	0.209	0.173	0.266	0.936	0.029	0.847

At the code level (the most granular ICD-10 layer), the flat PLM-ICD/HiLAT baseline achieved the highest **micro-F1 score of 0.336**, indicating stronger overall instance-level performance. This result stems largely from its independence across labels and its simpler training dynamics, which allow the model to prioritize frequently occurring codes. The hierarchical XR-LAT model, while showing a lower micro-F1 (**0.209**), demonstrated a markedly superior **macro-AUC of 0.847** compared to the baseline (0.793), confirming its advantage in rare label detection and probability ranking consistency.

The performance gap in micro-level metrics reflects a known trade-off: hierarchical models tend to sacrifice recall on frequent labels due to inter-level error propagation. Once a parent node (e.g., Chapter or Category) is incorrectly predicted, its descendants (fine-grained codes) cannot be activated. Nevertheless, XR-LAT’s higher AUC suggests more reliable confidence calibration, particularly for low-frequency or tail codes.

5.2 Layer-wise Results of XR-LAT

Table 3: Layer-wise results of XR-LAT on CodiEsp-D (Test).

Level	# Labels	micro-F1	macro-F1	macro-AUC
Chapter (L0)	25	0.735	0.519	0.865
Category (L1)	809	0.229	0.061	0.825
Code (L2)	1767	0.209	0.029	0.847

The hierarchical decomposition of XR-LAT reveals a gradual performance decline from high-level Chapters (L0) to fine-grained Codes (L2), which is expected in ICD-10 coding tasks. The **Chapter-level micro-F1 of 0.735** indicates that the model captures broad diagnostic categories effectively, whereas deeper levels require fine-grained contextual reasoning that suffers under limited data availability.

Interestingly, the **macro-AUC** remains relatively stable across layers ($0.865 \rightarrow 0.847$), demonstrating that even when discrete predictions degrade, probabilistic separability between positive and negative labels remains consistent.

6 Conclusion

We reproduced and analyzed the PLM-ICD/HiLAT baseline and XR-LAT architectures on the CodiEsp-D dataset. The findings replicate the characteristic trade-off between micro-F1 and macro-AUC, confirming the feasibility of hierarchical Transformers on small, multilingual medical datasets. Future work should explore beam-based hierarchical inference, adaptive threshold optimization, and domain-adaptive pretraining to enhance robustness and interpretability for real-world ICD coding.

References

- [1] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. <https://doi.org/10.18653/v1/N19-1423>
- [2] Liu, L., Perez-Concha, O., Nguyen, A., Bennett, V., & Jorm, L. (2023). Automated ICD coding using extreme multi-label long text transformer-based models. *Artificial Intelligence in Medicine*, 144, 102662. <https://doi.org/10.1016/j.artmed.2023.102662>
- [3] Ridnik, T., Ben-Baruch, E., Noy, A., & Zelnik, L. (2021). Asymmetric loss for multi-label classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. <https://doi.org/10.1109/ICCV48922.2021.00385>